Temperature

Temperature is the hotness or coldness of a substance. The temperature of a particular place changes based on the factors like latitude, altitude, season, time of day, cloud cover, wind, location of the sea etc. Temperature is measured utilizing thermometers. They are Centigrade, Fahrenheit, Wet Bulb, Dry Bulb and Tele thermometers.

In Centigrade thermometer, the temperature of melting ice is taken as 0^0 C and the temperature of boiling water as 100^0C. In Fahrenheit thermometer, 32^0F represents the freezing point and 212^0F represents the boiling point of water. Wet Bulb and Dry Bulb thermometers are used to measure humidity in temperature. Tele thermometer is used to record temperature continuously within and outside the building. Thermograph produces a continuous trace of the changing temperature of a place.

Pressure

Pressure is defined as the weight of a unit column of air at any place. The barometer is an instrument to measure atmospheric pressure. While both Barometer and Aneroid Barometer measures atmospheric pressure above and below sea level, Altimeter measures air pressure at a height above sea level. Barograph provides a continuous record of air pressure. The unit of pressure is measured in millibars (mb). Globally, the pressure determines the wind and weather pattern. High pressure and low pressure are highly related with the increase or decrease in temperature.

Wind

The wind is moving air over the earth's surface. The air moves from high pressure to low pressure either vertically or horizontally. Wind vane records the direction of the wind and points out from which direction it blows. Cup Anemometer measures wind velocity, which is expressed in knots.(1.852 km speed) Wind Sock is a device that indicates the origin of wind direction and speed.

Humidity

The amount of water vapour present in the atmosphere is termed as Humidity. It concentrates in the lower layer of the atmosphere. It varies from place to place and time to time. It may be classified as absolute, specific and relative humidity. Hygrometer is the instrument that measures the relative humidity. Hygrograph is an automatic instrument showing the change in relative humidity. Hygrothermograph records both humidity and temperature.

Cloudiness

Cloudiness indicates the amount of sky covered by clouds. Clouds are present in the lower atmosphere at varying altitude. They are varying in size and shape. The clouds are classified as high, middle and low clouds. Ceilometer is an instrument that uses a laser to determine the height of the cloud base. Ceiling projectors measure the height of the base of clouds above the ground. Cloud mirror measures the percentage of cloud cover. An Okta is a unit of measurement to describe the amount of cloud cover. The measurement of 0 Okta indicates clear sky and 8 Okta denotes completely overcast sky.

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